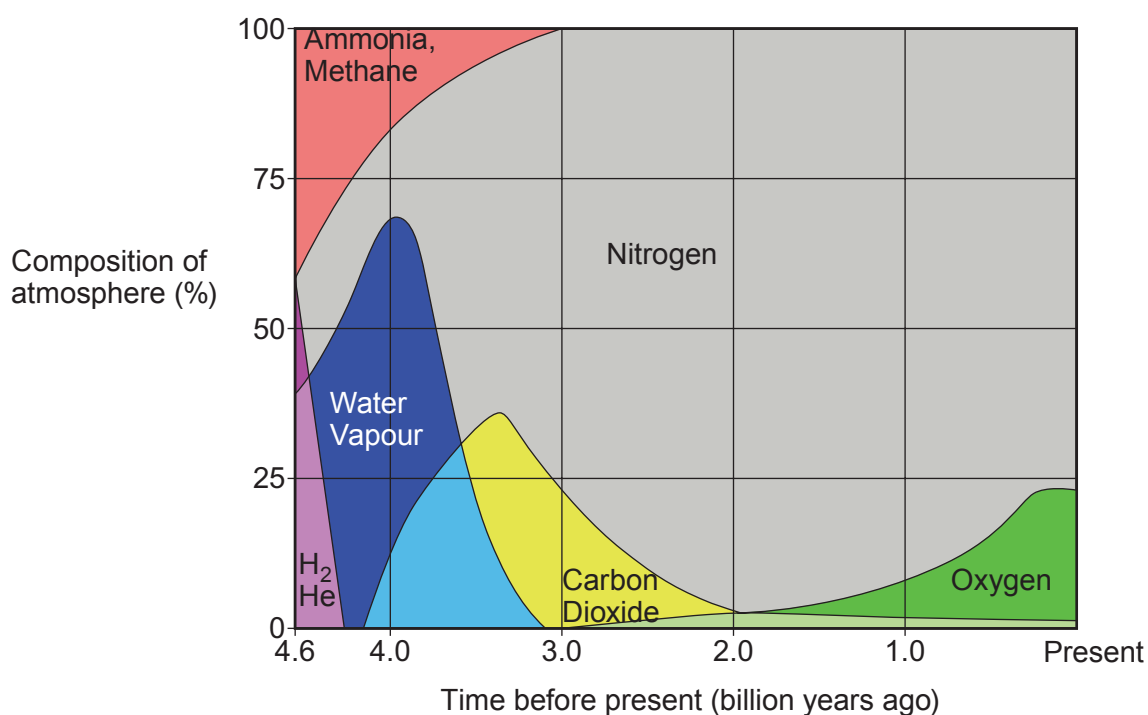


Examiner
only

7. The graph below shows how the composition of the atmosphere has changed over time, since the formation of the Earth 4.6 billion years ago.



- (a) Describe and explain how the composition of the atmosphere has changed over time.

[6 QER]

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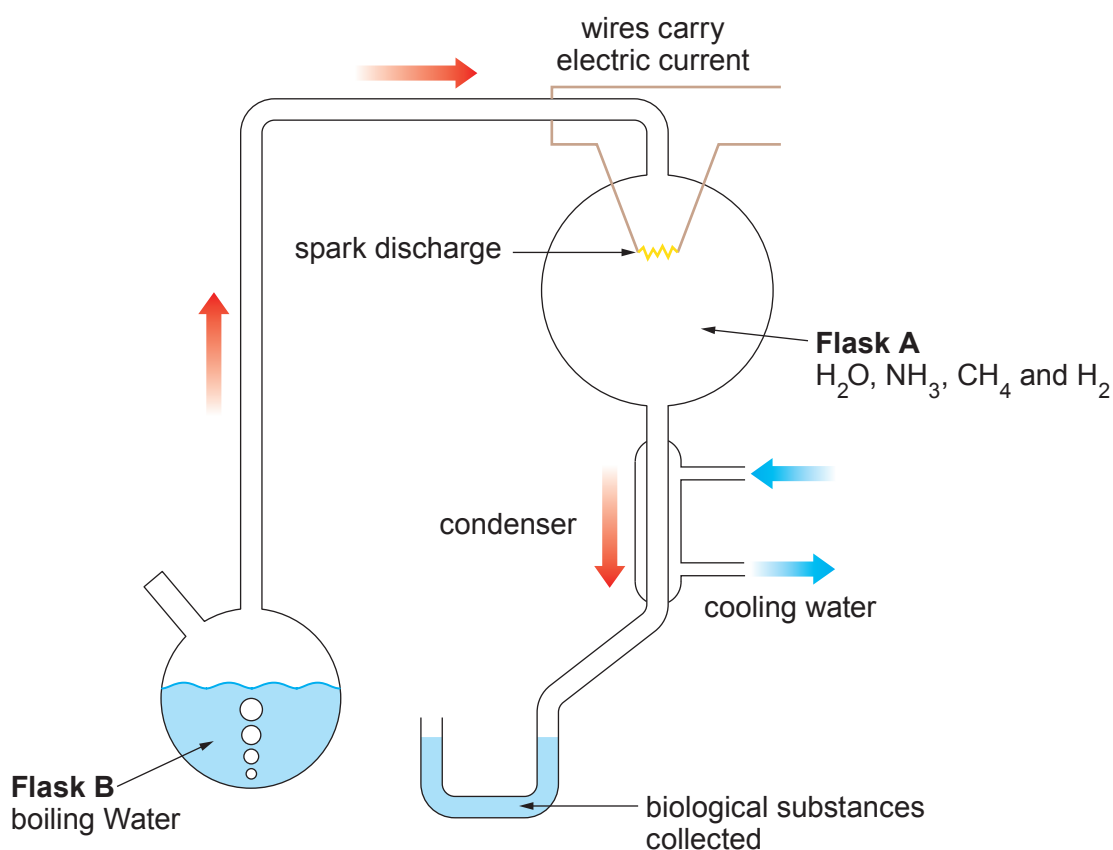
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- (b) In 1952 two scientists, Stanley Miller and Harold Urey, carried out experiments to see if the building blocks of common biological substances like carbohydrates (e.g. $C_6H_{12}O_6$), fatty acids (e.g. $CH_3(CH_2)_{17}COOH$), and amino acids (e.g. H_2NCH_2COOH) could be formed in the atmosphere of early Earth.

The scientists set up the apparatus shown below.



- (i) Explain why Miller and Urey, chose the gases in **Flask A**.

[2]

- (ii) Use the information in the diagram to state **two** assumptions that Miller and Urey made about the early Earth other than the composition of the atmosphere.

[2]

END OF PAPER